

Output 1

```
arnav@LAPTOP-BN8992KC:~/OS$ nano week2a.c
arnav@LAPTOP-BN8992KC:~/OS$ nano week2a.c
arnav@LAPTOP-BN8992KC:~/OS$ gcc week2a.c -o week2a
arnav@LAPTOP-BN8992KC:~/OS$ ./week2a
File copied successfully.
arnav@LAPTOP-BN8992KC:~/OS$ |
```

Program1

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;
    src = open("input.txt", O_RDONLY);
    if(src < 0){
        printf("Cannot open source file\n");
        return 1;
    }
    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }
    close(src);
    close(dest);
    printf("File copied successfully.\n");
    return 0;
}
```

Program 2

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

Output

```
arnav@LAPTOP-BN8992KC:~/OS$ nano week2b.c
arnav@LAPTOP-BN8992KC:~/OS$ gcc week2b.c -o week2b
arnav@LAPTOP-BN8992KC:~/OS$ ./week2b
Enter Linux command: ps
Parent PID : 1338
Child PID : 1339
  PID TTY          TIME CMD
  334 pts/0        00:00:00 bash
 1338 pts/0        00:00:00 week2b
 1339 pts/0        00:00:00 ps
Child process completed.
arnav@LAPTOP-BN8992KC:~/OS$ |
```